1.Create a SQL script to insert a new order into an Orders table, then use COMMIT to save the transaction and verify that the new order persists after reconnecting to the database.
Answer:

```
INSERT INTO Orders (order_id, customer_id, order_date, total_amount)
VALUES (101, 5, CURRENT_DATE, 450.00);
COMMIT;
SELECT * FROM Orders WHERE order_id = 101;
```

2.Simulate a Zomato-style food order process: insert two new items into an OrderItems table, then use ROLLBACK to undo the changes before committing. Check that no new items remain in the table after rollback.
Answer:

```
INSERT INTO OrderItems (order_id, customer_id, order_date, total_amount)VALUES
(1, 501, 'Paneer Tikka', 2, 300.00);
(2, 501, 'Butter Naan', 4, 160.00);
ROLLBACK;
SELECT * FROM OrderIterms WHERE order_id = 502;
```

3.In a Flipkart-like shopping cart scenario, use SAVEPOINT to mark a point after adding a product to the Cart table, add another product, then use ROLLBACK TO SAVEPOINT to undo only the last addition. Show the final contents of the Cart table.
Answer:

```
INSERT INTO Cart(cart_id, user_id, product_name, price) VALUES
        (1, 10, 'Wireless Mouse', 799.00);
SAVEPOINT first_product;
INSERT INTO Cart (cart_id, user_id, product_name, price) VALUES
        (2, 10, 'Bluetooth Speaker', 1499.00);
ROLLBACK TO SAVEPOINT after_first_product;
SELECT * FROM Cart WHERE user_id = 10;
```

4.Write a trigger on a Wallet table that automatically deducts the purchase amount from the user's balance whenever a new transaction is inserted, similar to how Paytm updates wallet balance after a payment.
Answer:

```
CREATE OR REPLACE TRIGGER trg_update_wallet_balance
        AFTER INSERT ON Transactions
        FOR EACH ROW
        BEGIN
                UPDATE Wallet SET balance = balance - :NEW.amount
                WHERE user_id = :NEW.user_id; END;
```

5.Use ChatGPT to generate a SQL test scenario where a trigger incorrectly updates a user's balance after an expense entry. Paste the scenario and your corrected trigger code, explaining how you fixed the error.

Hint: Ask ChatGPT for a buggy trigger example and a test case that exposes the bug.
Answer:

BUGGY CODE:

```
CREATE OR REPLACE TRIGGER trg_buggy_wallet
AFTER INSERT ON Transactions
FOR EACH ROW
BEGIN
   UPDATE Wallet
   SET balance = balance + :NEW.amount;
END;
/
```

CORRECTED CODE:

```
CREATE OR REPLACE TRIGGER trg_fixed_wallet
AFTER INSERT ON Transactions
FOR EACH ROW
BEGIN
   UPDATE Wallet
   SET balance = balance - :NEW.amount
   WHERE user_id = :NEW.user_id;
END;
/
```

The code was corrected by correcting the operators from + to -.
WHERE clause was added as only new user’s accounts get altered.